

AI-Based Industrial Maintenance & Process Optimization

FactoryOps Predictive Maintenance Platform

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1. Introduction & Problem Statement

INTRODUCTION

Industrial assets continuously generate temperature, pressure, vibration, RPM, humidity, oil level, voltage, current and power telemetry.

The challenge is converting this high-volume operational data into timely maintenance intelligence.

AI-BASED APPROACH

The AI-based approach learns patterns in telemetry, flags abnormal behaviour, estimates machine health and supports failure-risk prediction.

This connects monitoring, analytics and maintenance actions in a single operational workflow.

PROBLEM STATEMENT

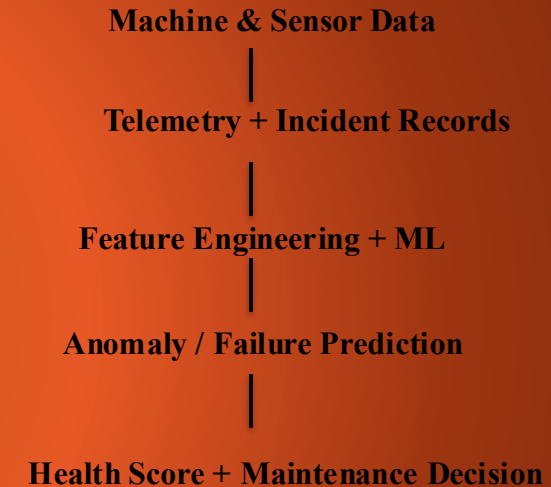
Unexpected failures interrupt production and can increase repair cost. Manual inspection alone makes continuous monitoring difficult. The system addresses this by turning telemetry and incident data into prioritized alerts and actionable maintenance information.

2. Objectives & System Architecture

OBJECTIVES

- **Monitor machine condition and telemetry**
- **Detect abnormal operating behaviour**
- **Predict potential failure risk**
- **Score machine health for prioritization**
- **Track incidents, work orders and maintenance cost**
- **Provide APIs and dashboard-based decision support**

SYSTEM ARCHITECTURE

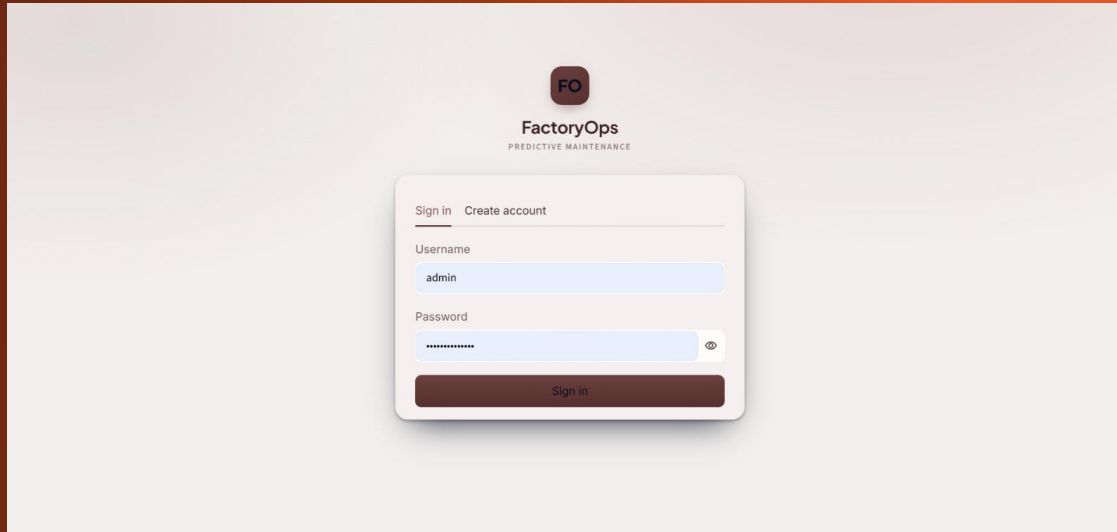


Presentation Layer

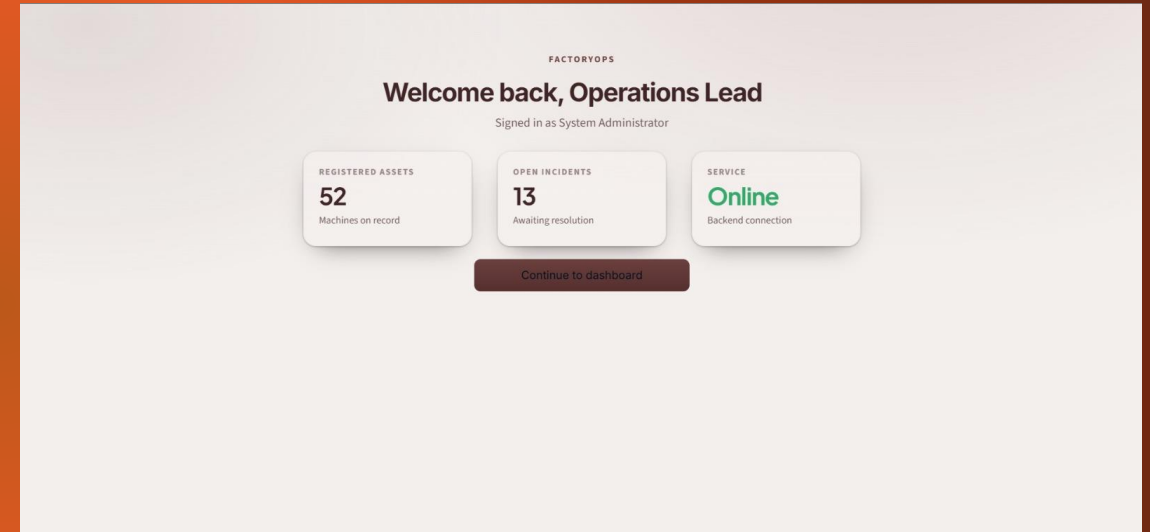
Backend APIs → Frontend Dashboard → Engineers

3. Frontend & Application Access

Secure entry point for operational users, with role-aware access to monitoring and maintenance functions.



Login / role-based access
Authenticates the user before operational modules are exposed.



Operations Lead welcome screen
Provides a clear route into machine monitoring and maintenance workflows.

4. Frontend — Dashboard

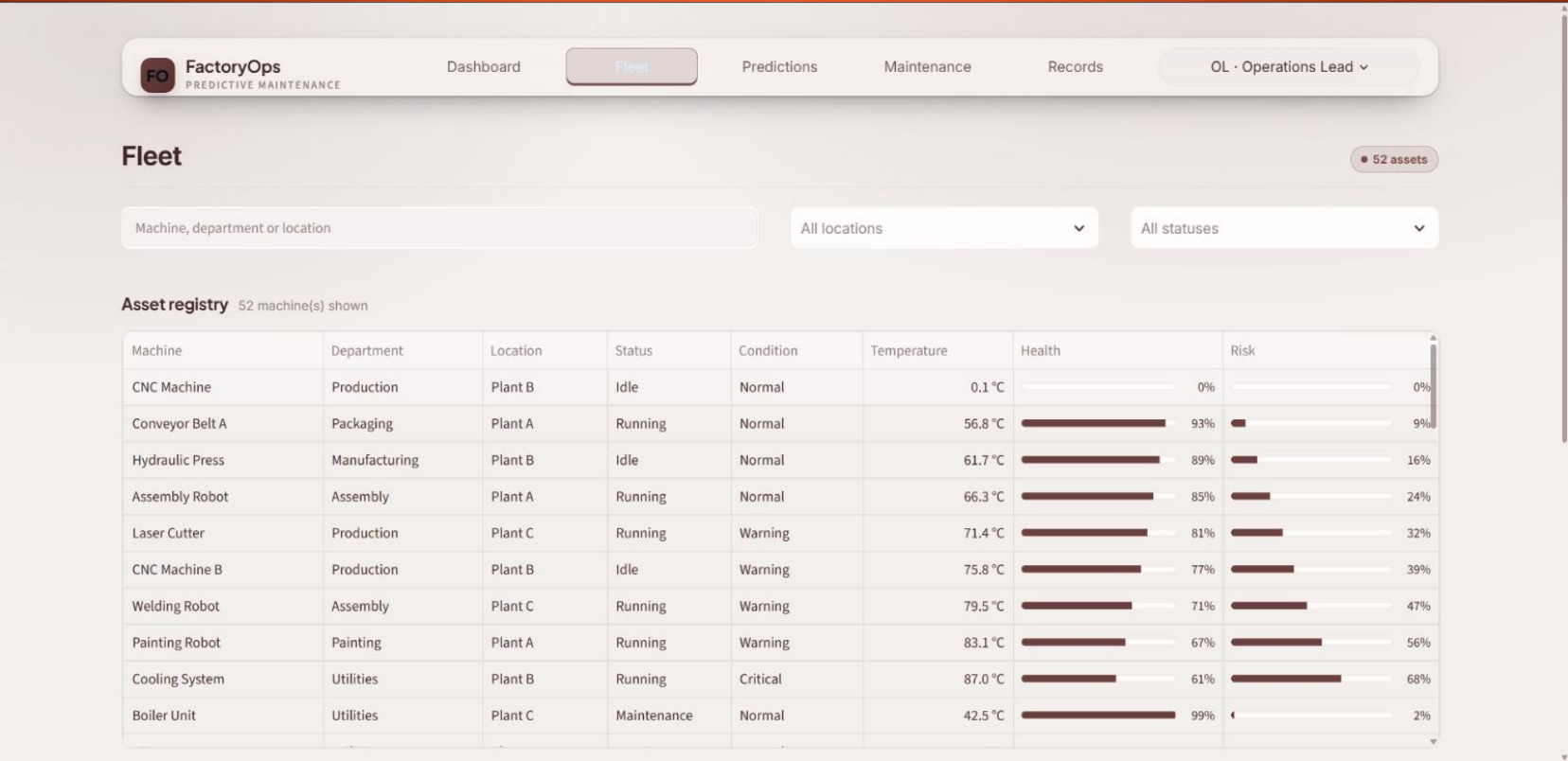
Central monitoring screen for factory operations -bringing machine status, telemetry trends, alerts and maintenance information together.



Dashboard role: Observe → Interpret → Prioritize → Act

5. Fleet & Machine Monitoring

Asset registry, machine condition and operational status. Fleet-level visibility helps teams identify which assets require attention first.



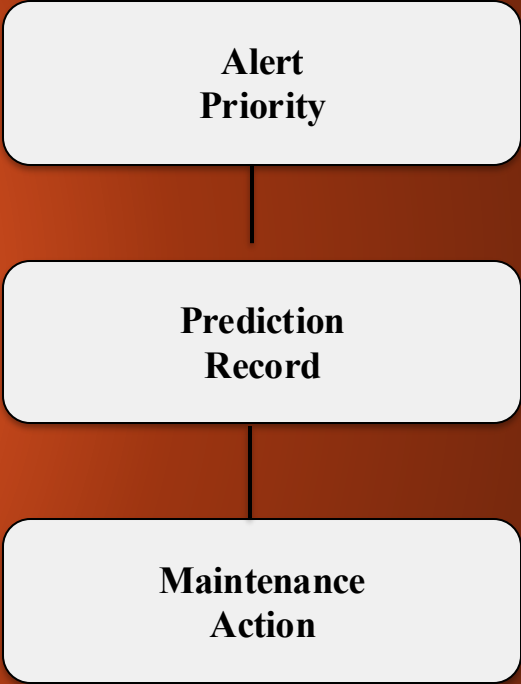
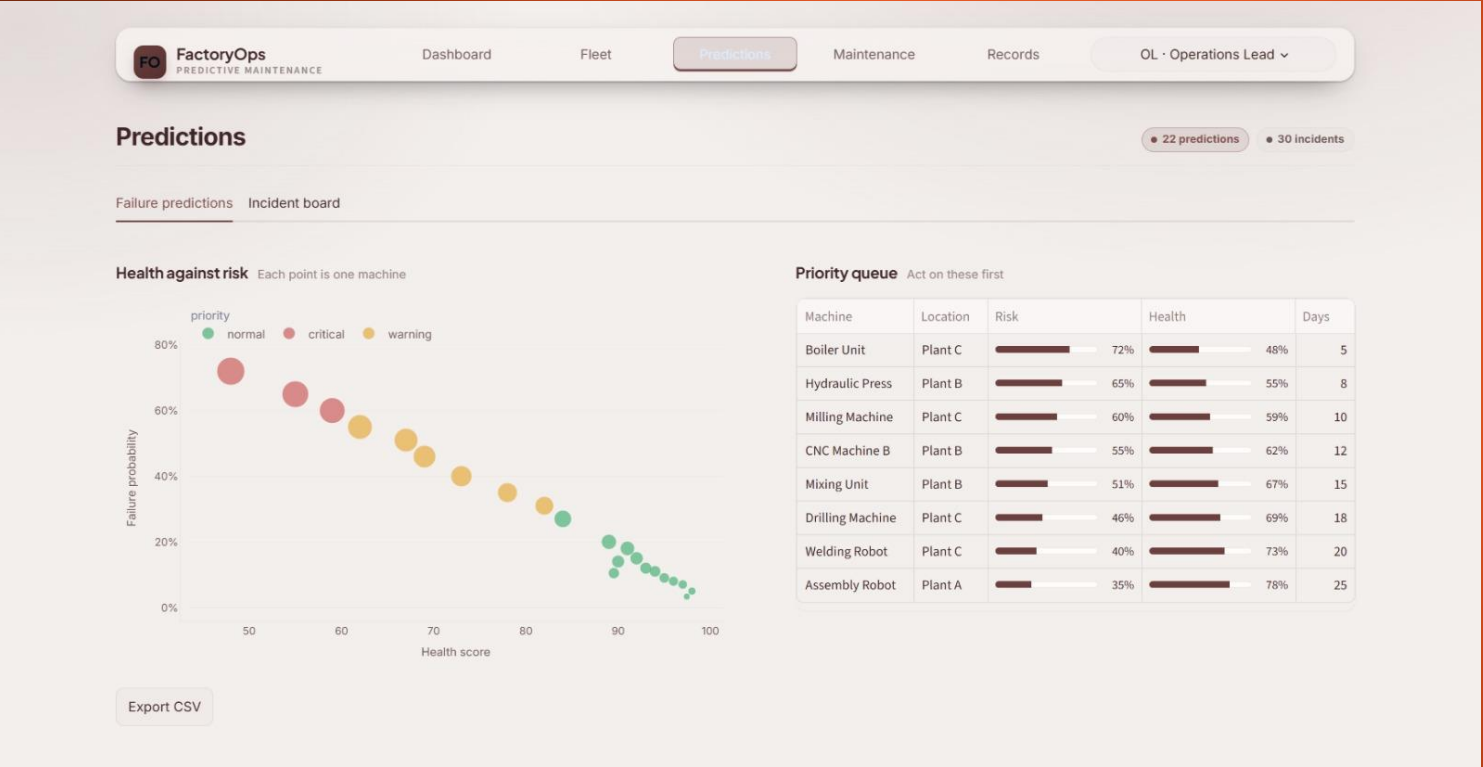
Asset Registry

Telemetry Monitoring

Health & Status

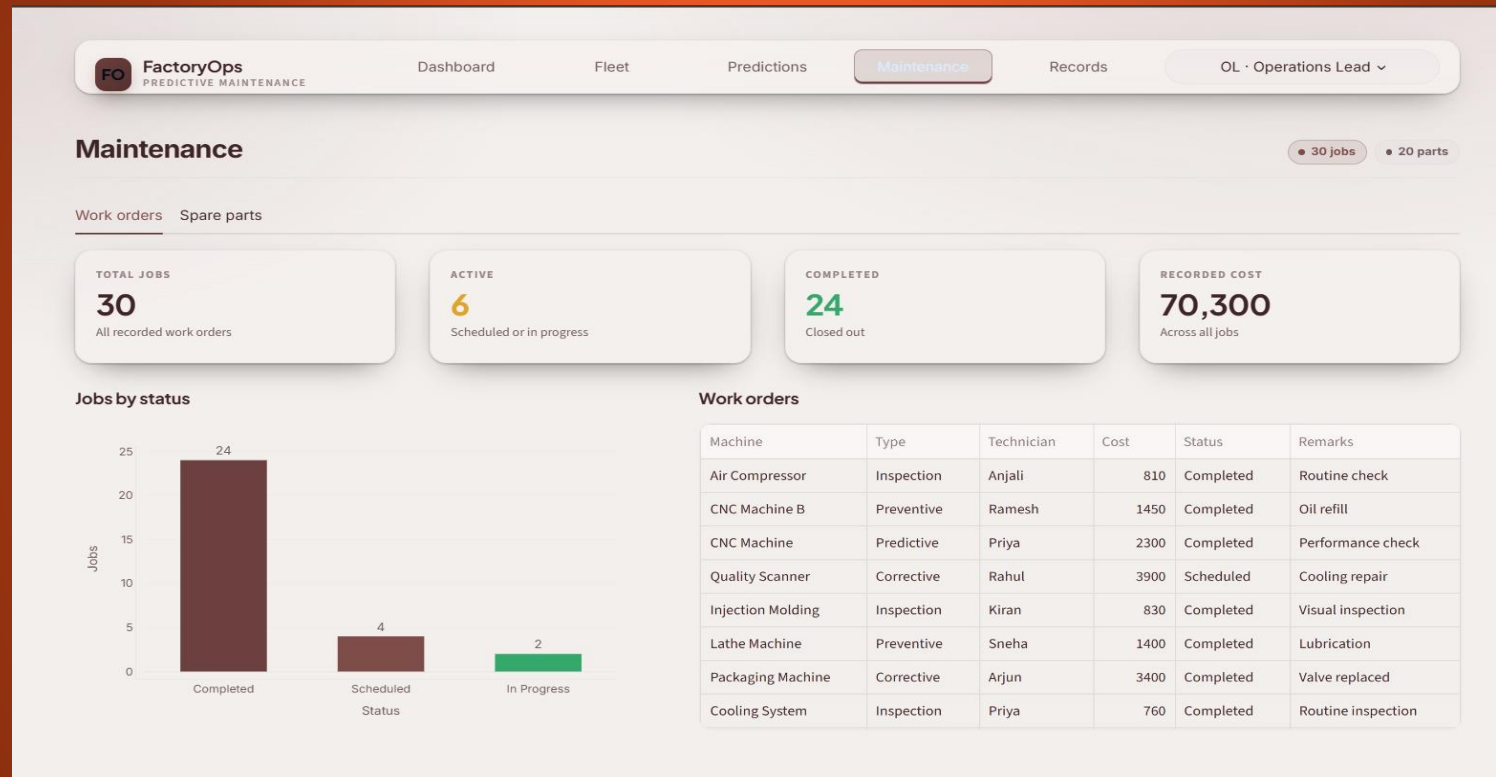
6. Predictions & Alerts

Failure-risk visualization converts model output into a priority queue so maintenance teams can focus on machines showing abnormal or elevated-risk behaviour.



7. Maintenance Management

Work orders connect detected maintenance needs with execution status and recorded cost, creating a traceable maintenance lifecycle.



Issue Identified

Work Order

Status + Cost Record

8. Records & Data Management

Operational records provide the data foundation for analytics: machines, telemetry, incidents, maintenance activities and prediction outputs.

The screenshot displays the 'FactoryOps PREDICTIVE MAINTENANCE' dashboard. The top navigation bar includes links for Dashboard, Fleet, Predictions, Maintenance, and Records (which is highlighted). A user profile 'OL · Operations Lead' is visible on the right. Below the navigation bar, the 'Records' section is active, with a 'Service online' status indicator. A sub-navigation bar lists categories: Machine, Telemetry, Maintenance, Incident, Inventory, Notification, and Sensor. The 'Machine' category is selected. Under 'Register a machine', there are input fields for 'Machine name' (with a 'Required' label), 'Department', 'Location', and a 'Status' dropdown menu currently set to 'Running'. An 'Add machine' button is positioned below the location field. The footer contains copyright information '© 2026 FactoryOps · Predictive Maintenance Platform', links for Documentation, API Reference, Support, and Privacy, the version 'v1.0.0', and another 'Service online' indicator.

Machines

Telemetry

Incidents

Maintenance

Predictions

9. Backend Overview

Layered backend architecture keeps API handling, validation, business logic, persistence and ML analytics modular and maintainable.

API LAYER

REST APIs
FastAPI + Uvicorn
Swagger documentation
Frontend communication

DATABASE

SQLite + SQLAlchemy
Machine data
Telemetry
Incidents & maintenance
Prediction records

SERVICES

Service Layer
Health / anomaly
Prediction
Maintenance
Process business rules

Backend flow:

Frontend request → API endpoint → Schema validation → Service logic → Database / ML service → Structured response.

ML integration: telemetry data is prepared and transformed into model-ready features, predictions are returned to the application, and results are exposed through APIs for dashboard consumption.

Python

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FastAPI

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SQLAlchemy

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SQLite

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Pandas

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NumPy

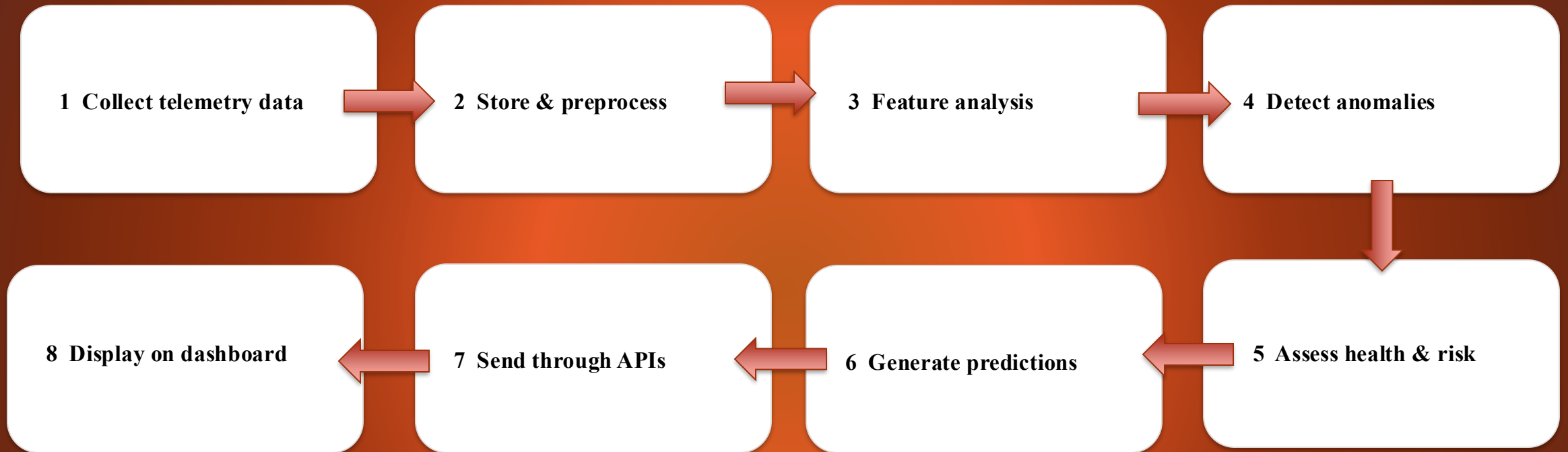
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Scikit-learn

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XGBoost

10. AI/ML Module & System Workflow



AI/ML pipeline: Telemetry → Preprocessing → Feature Engineering → Anomaly Detection → Health Assessment → Failure-Risk Prediction → API Delivery → Dashboard Action

Anomaly Detection: identify deviations from expected machine behaviour

Failure Prediction: estimate potential failure risk from learned patterns

Health Scoring: convert signals into an actionable machine condition view

11. Results & Advantages

RESULTS

The platform centralizes machine health visualization, telemetry trends, alerts, incidents and maintenance/prediction records.

- **Early identification of abnormal machine behaviour**
- **Reduced dependency on continuous manual monitoring**
- **Better maintenance prioritization through risk information**
- **Potential reduction in unexpected downtime**
- **Improved visibility of machine health and operational data**
- **Data-driven support for industrial decisions**

OPERATIONAL OUTPUT

Operational teams receive a single interface for reviewing machine condition, identifying abnormal behaviour and prioritizing maintenance information.

USE CASES

- **Manufacturing plants**
- **Production lines**
- **Power plants**
- **Automotive industries**
- **Process industries**

12. Future Scope & Conclusion

FUTURE SCOPE

- **Real-time IoT sensor integration and streaming telemetry**
- **Advanced deep-learning models for complex failure patterns**
- **Remaining Useful Life (RUL) prediction**
- **Automated maintenance scheduling**
- **Cloud deployment and scalable data infrastructure**
- **Digital twin integration and closed-loop process optimization**
- **AI agents for process and maintenance assistance**

CONCLUSION

The system combines AI/ML, industrial telemetry, FastAPI services, structured records and an interactive Streamlit interface.

It supports machine health assessment, anomaly detection, failure-risk prediction and maintenance tracking.

Goal: move from reactive maintenance toward predictive, data-driven maintenance and improved process reliability.

THANK YOU!

Questions & Discussion

Team D | FactoryOps Predictive Maintenance Platform
